Smart Process Management Agent

Agentic AI Based Operating System Optimization System

Abstract

This project presents a Smart Process Management Agent that monitors system resources and performs automated optimization actions using agentic AI concepts. The system observes CPU and memory usage, analyzes running processes, makes decisions based on predefined conditions, and performs actions such as alerts or process optimization. The project demonstrates operating system interaction, automation, and intelligent decision-making using Python and the psutil library.

Introduction

Modern operating systems execute multiple processes simultaneously, which can cause high CPU and memory usage. Existing monitoring tools provide information but require manual user actions. This project introduces an intelligent automation system capable of monitoring system performance and making optimization decisions automatically.

Problem Statement

Many users experience system slowdowns caused by unnecessary or resource-heavy background applications. Existing monitoring tools require users to manually analyze and optimize the system.

This project addresses this problem by developing a Smart Process Management Agent.

Objectives

- Monitor CPU and memory usage in real time
- Detect high resource-consuming processes
- Implement agentic AI decision-making
- Automate optimization actions
- Demonstrate operating system process management

Methodology

The system follows the agentic AI workflow:

Observe → Analyze → Decide → Act

1. Observe system processes and resource usage.
2. Analyze CPU and memory thresholds.
3. Decide whether optimization is required.
4. Execute actions such as alerts or process termination.

System Architecture

Running Processes → Monitoring Module → Decision Engine → Action Handler → Optimization Result

Tools and Technologies

• Python
• psutil library
• Streamlit
• VS Code

Implementation

The project contains three main modules:

1. Monitoring Module:
Reads CPU usage, RAM usage, and active processes.

2. Decision Engine:
Analyzes resource usage and determines optimization actions.

3. Action Handler:
Generates alerts or performs process optimization actions.

Expected Output

The system continuously monitors system resources and detects abnormal CPU or memory usage. When thresholds are exceeded, the agent generates optimization suggestions or performs actions automatically.

Advantages

• Automated system monitoring
• Intelligent decision-making
• Lightweight implementation
• Real-time optimization support

Limitations

• Limited to local machine monitoring
• Basic rule-based intelligence
• Requires careful process management

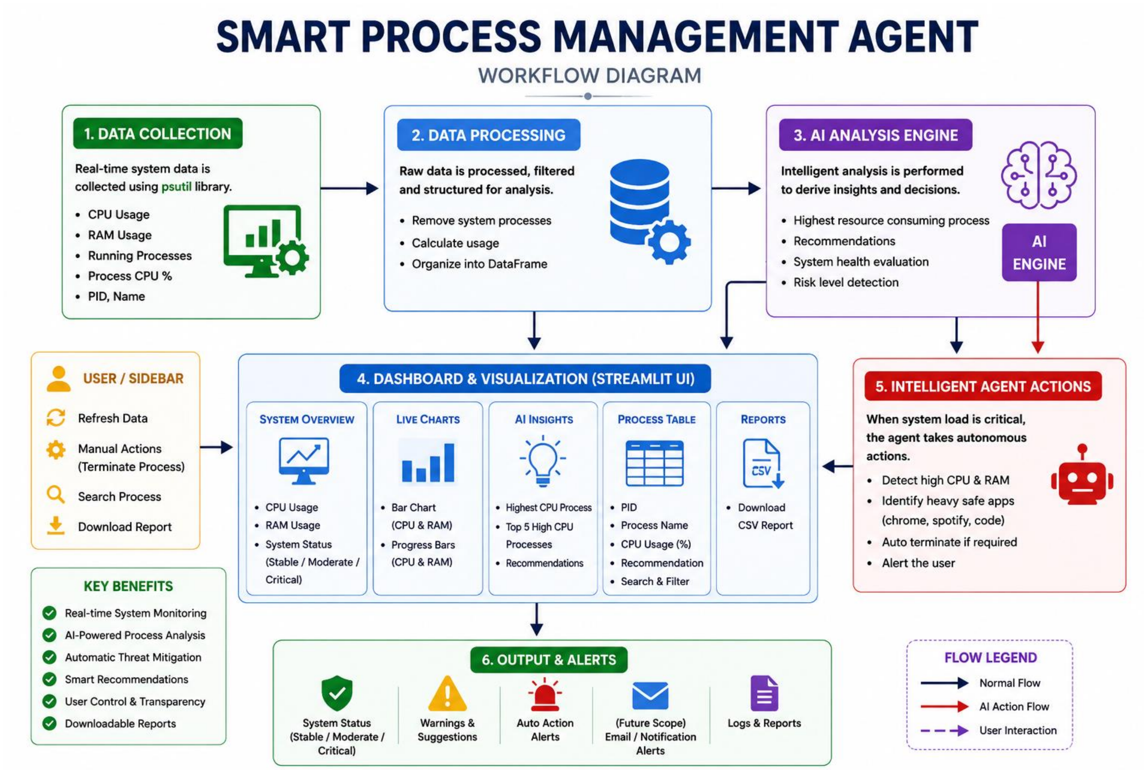
Future Improvements

• Machine learning-based optimization
• Advanced dashboards
• Predictive system monitoring

4-Week Project Plan

Week 1: Research & Environment Setup
Week 2: Monitoring & Decision Module Development
Week 3: Action Handler & System Integration
Week 4: Testing, UI Development & Final Presentation

Project Diagram



Conclusion

The Smart Process Management Agent successfully demonstrates the integration of agentic AI concepts with operating system process management. The system performs monitoring, decision-making, and optimization actions automatically while remaining lightweight and practical for educational purposes.